Journal of Industrial Intelligence
https://www.acadlore.com/journals/JII

Hybrid Quantum–Classical Framework for Fuzzy 0–1 Integer Linear Programming: α-Level Trade-off Analysis via Quadratic Unconstrained Binary Optimization Reformulation

Hamed Fazlollahtabar[1*], Hadi Gholizadeh[2]

[1] Department of Industrial Engineering, School of Engineering, Damghan University, 36716 Damghan, Iran
[2] Département de Génie Mécanique, Université Laval, G1V 0A6 Québec, Canada

[*] Correspondence: Hamed Fazlollahtabar (h.fazl@du.ac.ir)

Received: 06-25-2025 **Revised:** 08-10-2025 **Accepted:** 08-25-2025

Citation: H. Fazlollahtabar and H. Gholizadeh, "Hybrid quantum–classical framework for fuzzy 0–1 Integer Linear Programming: α-level trade-off analysis via Quadratic Unconstrained Binary Optimization reformulation," *J. Ind Intell.*, vol. 3, no. 3, pp. 172–188, 2025. https://doi.org/10.56578/jii030304.

Abstract: This paper addressed the computational challenges of solving large-scale fuzzy 0–1 Integer Linear Programming (ILP) problems by proposing a Fuzzy-Hybrid Quantum-Classical Optimization Algorithm (F-HQCOA). The novel approach employed α-cut based transformations to convert fuzzy ILP models into a family of crisp Quadratic Unconstrained Binary Optimization (QUBO) problems, which were then solved using hybrid quantum-classical techniques. Unlike multi-objective formulations, the proposed framework performed a parametric analysis over α-levels to generate a spectrum of solutions that reflected trade-offs between objective optimality and feasibility satisfaction under uncertainty. The degree of satisfaction (μ) was adopted as an evaluation metric to assess solution quality rather than as an independent optimization objective. The method was evaluated on fuzzy Multi-Dimensional Knapsack and Set Covering problems. Results showed that the proposed approach could efficiently handle larger instances and provided a diverse set of solutions across α-levels. While quantum-assisted methods demonstrated reduced Time-to-Solution (TTS) in certain settings, comparisons were presented with careful consideration of differences in computational paradigms. The proposed framework offers a practical decision-support tool for optimization under uncertainty and establishes a foundation for future extensions toward true multi-objective fuzzy quantum optimization.

Keywords: Fuzzy optimization; Quantum computing; Integer programming; Quantum annealing; Quadratic Unconstrained Binary Optimization; Uncertainty; Fuzzy-Hybrid Quantum-Classical Optimization Algorithm

1 Introduction

Integer Linear Programming (ILP) represents a fundamental class of optimization problems with wide-ranging applications in logistics, finance, telecommunications, and resource allocation. The 0–1 ILP variant, where variables are restricted to binary values $\{0, 1\}$, is particularly significant as it models numerous combinatorial optimization problems with substantial practical importance. However, traditional 0–1 ILP formulations assume precise knowledge of all parameters, which rarely aligns with real-world conditions where costs, demands for resources, and constraints are often imprecise or vaguely defined [1].

Fuzzy set theory, introduced by Zadeh [2], provides a mathematical framework for modeling this uncertainty through membership functions and fuzzy numbers. While fuzzy optimization methods have been developed over decades, they face substantial computational challenges when applied to large-scale combinatorial problems due to their exponential worst-case complexity. Concurrently, quantum computing has emerged as a promising paradigm for tackling combinatorial optimization, with approaches like quantum annealing (QA) and the Quantum Approximate Optimization Algorithm (QAOA) showing potential for speedup on certain problem classes [3].

Modern industrial systems feature growing complexity, dynamism and uncertainty. Decision-making under such conditions often relies on imprecise information stemming from measurement errors, subjective judgement, prediction inaccuracies, and intrinsic process variability. Such uncertainty appears widely in industrial intelligence applications.

Production Resource Allocation: Manufacturing systems should allocate limited resources (machines, labor, and materials) across competing production activities. Demand forecasts, processing time, and resource availability are rarely known with certainty, thus necessitating optimization under fuzzy parameters. Production managers often express constraints such as “approximately 100 units of raw material are available” or “the processing time is about 5 hours”, which naturally fit the fuzzy paradigm.

Manufacturing Scheduling: Problems of job shop scheduling and flow shop scheduling involve fuzzy processing time, fuzzy due dates, and fuzzy machine availability. The objective often involves minimizing fuzzy makespan or maximizing fuzzy customer satisfaction, thus turning fuzzy ILP into a natural modeling framework.

Supply Chain Optimization: Supply chain networks face uncertainties in demand, supplier reliability, transportation costs, and lead time. Fuzzy optimization enables robust decision-making that accounts for these uncertainties while maintaining operational efficiency.

Energy Management Systems: Power systems optimization involves fuzzy load forecasts, fuzzy generation capacities, and fuzzy transmission constraints. The integration of renewable energy sources introduces additional uncertainty, rendering fuzzy optimization essential for reliable energy management.

Healthcare Resource Management: Hospital scheduling, patient flow optimization, and medical resource allocation involve imprecise parameters such as uncertain patient arrival time, variable durations of treatment, and unpredictable resource requirements.

Portfolio Optimization: Financial decision-making under uncertainty involves fuzzy returns, fuzzy risks, and fuzzy investment constraints, requiring sophisticated optimization tools that could handle imprecision.

The proposed Fuzzy-Hybrid Quantum-Classical Optimization Algorithm (F-HQCOA) framework dealt with these industrial challenges by providing a scalable and quantum-assisted solution methodology that could handle the inherent uncertainty of real-world industrial problems, while generating a spectrum of solutions to support risk-aware decision-making.

Despite these advancements, a significant gap exists in the literature regarding the integration of fuzzy optimization methodologies with quantum computing approaches. The main contributions of this work are summarized as follows:

1. A formal α-cut-based transformation framework for converting fuzzy 0–1 ILP problems into a family of deterministic Quadratic Unconstrained Binary Optimization (QUBO) models;
2. A F-HQCOA that solves each α-level problem efficiently using quantum optimization techniques;
3. A systematic parametric analysis framework that generates a spectrum of solutions across α-levels, enabling evaluation of trade-offs between feasibility satisfaction and objective performance;
4. An empirical evaluation demonstrating scalability and solution diversity compared to classical fuzzy optimization methods;
5. A discussion of the limitations of current quantum approaches and the potential for extending the framework toward multi-objective fuzzy optimization.

Notably, the proposed framework differs from classical Pareto-based multi-objective optimization. It employs an α-parametric method generating deterministic problems for each α, and solutions over varying α illustrate the trade-off between feasibility satisfaction and objective value. This distinction prevents misinterpretation against conventional multi-criteria optimization.

The paper proceeds as follows. Section 2 reviews relevant studies on fuzzy optimization and quantum computing. Section 3 outlines the theoretical framework and Section 4 specifies F-HQCOA. Section 5 describes experimental configurations, whose outcomes are analyzed in Section 6. Section 7 discusses implications and limitations, and Section 8 concludes with future research.

2 Literature Review

Recent advances in fuzzy integer programming have explored multi-objective formulations, interactive decision-making approaches, and metaheuristic solution techniques [4], [5]. In particular, multi-objective fuzzy optimization has been extensively studied using Pareto-based methods, weighted aggregation, and fuzzy goal programming.

In parallel, significant progress has been made in quantum optimization, particularly in QUBO formulations for combinatorial problems [6]. Hybrid quantum–classical algorithms such as QAOA and QA have been applied to many Nondeterministic Polynomial (NP)-hard problems, including scheduling, routing, and portfolio optimization [7], [8].

The fusion of fuzzy optimization and quantum computing remains underexplored. Existing deterministic approaches neglect uncertainty in quantum optimization. This work bridges the gap by integrating fuzzy modeling into QUBO-based quantum optimization.

2.1 Fuzzy Integer Programming

Fuzzy optimization approaches have evolved significantly since Zadeh’s pioneering work on fuzzy sets [2]. As an early contribution, Zimmermann [9] introduced fuzzy linear programming with multiple objective functions, while Verdegay [10] developed fundamental duality results. Traditional fuzzy optimization methods could be broadly categorized into possibilistic programming, fuzzy stochastic programming, and robust fuzzy programming approaches.

A fundamental challenge in fuzzy optimization involves handling imprecise parameters without dramatically increasing computational complexity. Recent advancements focused on granular differentiability (gr-differentiability) concepts to address limitations of earlier Hukuhara-based differentiation approaches. González-Rodelas et al. [11] proposed gr-differentiability to overcome issues such as increasing support closure length and unnatural behavior in modeling phenomena. This approach maintains a single fuzzy differential equation without multiple solution forms, to provide significant advantages for optimization under uncertainty.

Specifically for fuzzy integer programming, researchers have developed specialized branch-and-bound algorithms, cutting plane methods, and metaheuristics tailored to handle fuzzy constraints and objectives [12]. However, these approaches continue to face scalability challenges with problem sizes encountered in practical applications, necessitating new computational paradigms.

2.2 Quantum Optimization

Quantum optimization has emerged as a promising approach for combinatorial problems, with two primary paradigms: QA and gate-based quantum algorithms like QAOA [3]. Quantum annealers, such as those produced by D-Wave Systems, are specialized quantum computers designed to find low-energy states of Ising Hamiltonians, which could be mapped to QUBO problems [6].

Recent research has demonstrated the potential of quantum computing for various combinatorial optimization problems, though applications to fuzzy optimization remain largely unexplored. Current literature primarily focused on deterministic problems, with limited attention to uncertain or fuzzy environments. This gap is particularly notable given the potential for quantum approaches to tackle the exponential complexity of fuzzy combinatorial optimization.

2.3 Research Gap

Our review identified a significant research gap at the intersection of fuzzy optimization and quantum computing. While both fields have advanced independently, few works have attempted to integrate fuzzy set theory with quantum optimization approaches. The existing literature on fuzzy integer programming primarily focused on classical solution methods, which are facing scalability limitations. Meanwhile, quantum optimization research has predominantly handled deterministic problems without considering the imprecision and uncertainty inherent in real-world optimization scenarios.

This paper bridged these distinct research streams by developing a unified framework that leveraged the strengths of both fuzzy optimization theory and quantum computing capabilities. Our approach enabled the solution of larger-scale fuzzy ILP problems than previously possible with classical methods alone, while providing decision-makers with a spectrum of solutions representing different trade-offs between optimality and feasibility satisfaction.

3 Theoretical Framework

3.1 Definition of the Problem

We considered a fuzzy 0–1 ILP problem with imprecise parameters in both the objective function and constraints. Formally, the problem is defined as:

$$\text{Maximize } \tilde{c}^T x$$

subject to,

$$Ax \precsim \tilde{b}, \quad x \in \{0, 1\}^n$$

where, $\tilde{c} = (\tilde{c}_1, \tilde{c}_2, \ldots, \tilde{c}_n)$ and $\tilde{b} = (\tilde{b}_1, \tilde{b}_2, \ldots, \tilde{b}_m)$ denote vectors of triangular fuzzy numbers (TFNs). The symbol $\precsim$ stands for "approximately less than or equal to".

Each TFN is expressed as (a^p, a^m, a^o), where a^p denotes the pessimistic value, a^m denotes the most likely value, and a^o denotes the optimistic value [13].

The membership function of a TFN $\tilde{a}$ = (a^p, a^m, a^o) was given by:

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & \text{if } x < a^p \\ \dfrac{x - a^p}{a^m - a^p} & \text{if } a^p \leq x < a^m \\ 1 & \text{if } x = a^m \\ \dfrac{a^o - x}{a^o - a^m} & \text{if } a^m < x \leq a^o \\ 0 & \text{if } x > a^o \end{cases}$$

Figure 1 shows a TFN $\tilde{A}$ = (a^p, a^m, a^o) with its membership function $\mu_{\tilde{A}}(x)$. Three α-cuts (α = 0.0, α = 0.5, α = 1.0) are shown, demonstrating how each cut yielded a specific crisp interval [a^L_α,a^R_α].

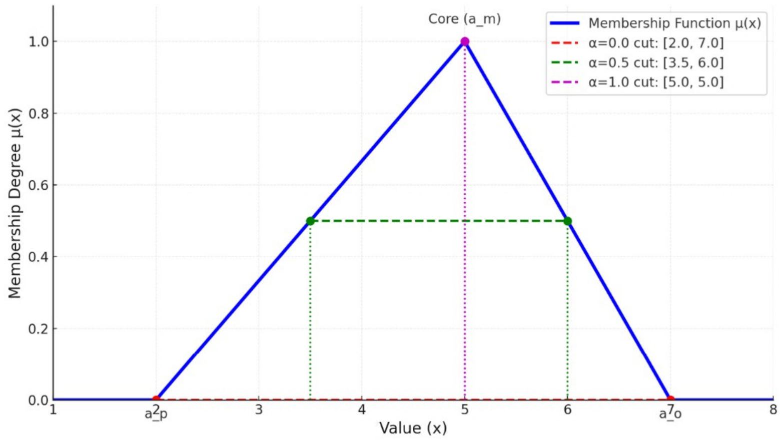

Figure 1. Triangular fuzzy number and α-cuts

3.2 α-Cut Based Transformation: Mathematical Foundation

To convert the fuzzy problem into a form amenable to quantum optimization, we employed the concept of α-cuts. For a fuzzy number $\tilde{a}$ and a specified $\alpha \in$ [0,1], the α-cut was defined as the crisp set:

$$[\tilde{a}]_\alpha = \left\{x \in X \,\middle|\, \mu_{\{\tilde{a}\}}(x) \geq \alpha\right\}$$

For a TFN $\tilde{a}$ = (a^p, a^m, a^o), the α-cut yielded the interval:

$$[\tilde{a}]_\alpha = \left[a^p + \alpha\left(a^m - a^p\right),\ a^o - \alpha\left(a^o - a^m\right)\right]$$

Rationale for Transformation Strategies: The transformation of fuzzy parameters to crisp values requires careful consideration of the optimization objective and constraint orientation.

In maximization problems, using the pessimistic value of the objective coefficients $c_j(\alpha) = c_j^p + \alpha\left(c_j^m - c_j^p\right)$ ensures that the resulting objective value is a conservative estimate of the achievable profit. This is appropriate for risk-averse decision-makers who prefer guaranteed lower bounds on profit. The parameter α controls the degree of optimism: at α = 0, the decision-maker accepts the most optimistic estimate c_j^p; at α = 1, the decision-maker accepts the most likely estimate c_j^m.

In maximization problems with resource constraints ($Ax \precsim \tilde{b}$), using the optimistic value of the capacity $b_i(\alpha) = b_i^o - \alpha(b_i^o - b_i^m)$ creates a stricter constraint. At α = 0, the constraint is most relaxed (using the optimistic bound b_i^o); at α = 1, the constraint is most restrictive (using the most likely bound b_i^m). This conservative transformation ensures that solutions satisfying the crisp constraints will satisfy the original fuzzy constraints to at least degree α.

Lemma 1 (Monotonicity of Feasible Region). For $\alpha_1 \leq \alpha_2$, the feasible region of the transformed crisp problem at α_2 is a subset of the feasible region at α_1.

Proof: For $\alpha_1 \leq \alpha_2$, we have $b_i(\alpha_2) \leq b_i(\alpha_1)$ since $b_i(\alpha) = b_i^o - \alpha(b_i^o - b_i^m)$ is decreasing in α. Therefore, the constraint $\sum_j a_{ij}x_j \leq b_i(\alpha_2)$ implies $\sum_j a_{ij}x_j \leq b_i(\alpha_1)$. Hence, the feasible region is monotonically decreasing with α.

This monotonicity property is critical for the parametric analysis. As α increases, the feasible region shrinks, leading to potentially lower objective values but higher confidence in feasibility.

3.2.1 Numerical example of α-cut transformation

To illustrate the α-cut transformation process, consider a small example with two variables and one constraint:

Fuzzy Problem:

$$\text{Maximize } \tilde{c}_1x_1 + \tilde{c}_2x_2$$

subject to,

$$\tilde{a}_1x_1 + \tilde{a}_2x_2 \precsim \tilde{b}$$

$$x_1, x_2 \in \{0, 1\}$$

where,

$$\tilde{c}_1 = (8, 10, 12), \quad \tilde{c}_2 = (12, 15, 18)$$

$$\tilde{a}_1 = (1.5, 2.0, 2.5), \quad \tilde{a}_2 = (2.0, 2.5, 3.0)$$

$$\tilde{b} = (4.0, 4.5, 5.0)$$

Transformation at α = 0.2
Objective coefficients:

$$c_1(0.2) = 8 + 0.2(10 - 8) = 8.4, c_2(0.2) = 12 + 0.2(15 - 12) = 12.6$$

Constraint Right-Hand Side (RHS):

$$b(0.2) = 5.0 - 0.2(5.0 - 4.5) = 4.9$$

Crisp problem:

Maximize $8.4x_1 + 12.6x_2$

subject to,

$$\big(1.5 + 0.2(2.0 - 1.5)\big)x_1 + \big(2.0 + 0.2(2.5 - 2.0)\big)x_2 \leq 4.9$$

Left-Hand Side (LHS) coefficients:

$$a_1(0.2) = 1.6, a_2(0.2) = 2.1$$

Feasible binary solutions:

$$(0, 0) : 0 \leq 4.9; (1, 0) : 1.6 \leq 4.9; (0, 1) : 2.1 \leq 4.9; (1, 1) : 3.7 \leq 4.9$$

Optimal solution:

$(1, 1)$ with objective value $8.4 + 12.6 = 21.0$

Transformation at α = 0.8
Objective coefficients:

$$c_1(0.8) = 8 + 0.8(10 - 8) = 9.6, c_2(0.8) = 12 + 0.8(15 - 12) = 14.4$$

Constraint RHS:

$$b(0.8) = 5.0 - 0.8(5.0 - 4.5) = 4.6$$

LHS coefficients:

$$a_1(0.8) = 1.5 + 0.8(2.0 - 1.5) = 1.9, a_2(0.8) = 2.0 + 0.8(2.5 - 2.0) = 2.4$$

Feasible binary solutions:

$$(0, 0) : 0 \leq 4.6; (1, 0) : 1.9 \leq 4.6; (0, 1) : 2.4 \leq 4.6; (1, 1) : 4.3 \leq 4.6$$

Optimal solution:

$(1, 1)$ with objective value $9.6 + 14.4 = 24.0$

Transformation at α = 1.0
Objective coefficients:

$$c_1(1.0) = 10, c_2(1.0) = 15$$

Constraint RHS:

$$b(1.0) = 4.5$$

LHS coefficients:

$$a_1(1.0) = 2.0, a_2(1.0) = 2.5$$

Feasible binary solutions:

$$(0, 0) : 0 \leq 4.5; (1, 0) : 2.0 \leq 4.5; (0, 1) : 2.5 \leq 4.5; (1, 1) : 4.5 \leq 4.5$$

Optimal solution:

$$(1, 1) \text{ with objective value } 10 + 15 = 25.0$$

Analysis of the Example: As α increases from 0.0 to 1.0:

The objective coefficients increase (from pessimistic to most likely values), resulting in higher objective values. The constraint RHS decreases (from optimistic to most likely values), rendering stricter constraint. However, in this example, the optimal solution (1,1) remains feasible for all α-levels.

The satisfaction degree μ for each solution was computed as the minimum membership grade:

For α = 0.2:

$$\mu = \min\Big(\mu_{\tilde{c}_1}(8.4), \mu_{\tilde{c}_2}(12.6), \mu_{\tilde{b}}(4.9)\Big) = \min(0.2, 0.2, 0.8) = 0.2$$

For α = 0.8:

$$\mu = \min\Big(\mu_{\tilde{c}_1}(9.6), \mu_{\tilde{c}_2}(14.4), \mu_{\tilde{b}}(4.6)\Big) = \min(0.8, 0.8, 0.2) = 0.2$$

The satisfaction degree μ is the minimum membership grade across all fuzzy constraints and the objective function for a given solution. Different solutions may have different μ values, and the solution's μ is not necessarily equal to the α-level used in the transformation.

This example demonstrated how different α-levels transformed the same fuzzy problem into different crisp problems, leading to a spectrum of solutions with varying objective values and feasibility satisfaction levels.

3.3 Quadratic Unconstrained Binary Optimization Formulation

Each crisp α-ILP problem was converted to a QUBO problem by incorporating constraints as penalty terms. The general QUBO form is:

$$\text{minimize: } x^T Q x$$

where, x is a vector of binary decision variables and Q is an $n \times n$ real-valued matrix.

Constraint i was incorporated using a quadratic penalty term:

$$P_i \left(\sum_j a_{ij} x_j - b_i(\alpha) \right)^2$$

The complete QUBO formulation for a given α became:

$$H(x, \alpha) = -\sum_j c_{j,\alpha} x_j + \sum_i P_i \left(\max\Big(0, \sum_j a_{ij} x_j - b_{i,\alpha}\Big) \right)^2$$

where, P_i are sufficiently large penalty weights that ensure constraint satisfaction in optimal solutions [14]. The α-Cut transformation rules are summarized in Table 1.

Table 1. α-cut transformation rules

Fuzzy Parameter	α-Cut Transformation	Interpretation
Objective coefficient $\tilde{c}_j$	$c_j(\alpha) = c_j^p + \alpha(c_j^m - c_j^p)$	Pessimistic value estimate
Constraint RHS $\tilde{b}_i$	$b_i(\alpha) = b_i^o - \alpha(b_i^o - b_i^m)$	Optimistic constraint bound

Note: RHS = Right-hand side.

4 Methodology: The Fuzzy-Hybrid Quantum-Classical Optimization Algorithm Framework

4.1 Overview of the Algorithm

The F-HQCOA integrated fuzzy set theory with quantum computation through a structured four-stage process.

α-Level Selection: Determine a set of α values $\{\alpha_1, \alpha_2, ..., \alpha_k\} \in$ [0,1] to explore different feasibility satisfaction levels.

Defuzzification: Transform the fuzzy problem into crisp QUBO problems for each α-level using the α-cut method.

Quantum Processing: Solve each QUBO problem using quantum algorithms (QA or QAOA).

Classical Refinement: Filter and refine solutions to ensure feasibility and evaluate satisfaction degrees.

The end-to-end workflow of the F-HQCOA, as shown in Figure 2, illustrates the four-stage process from fuzzy input to solutions across α-levels.

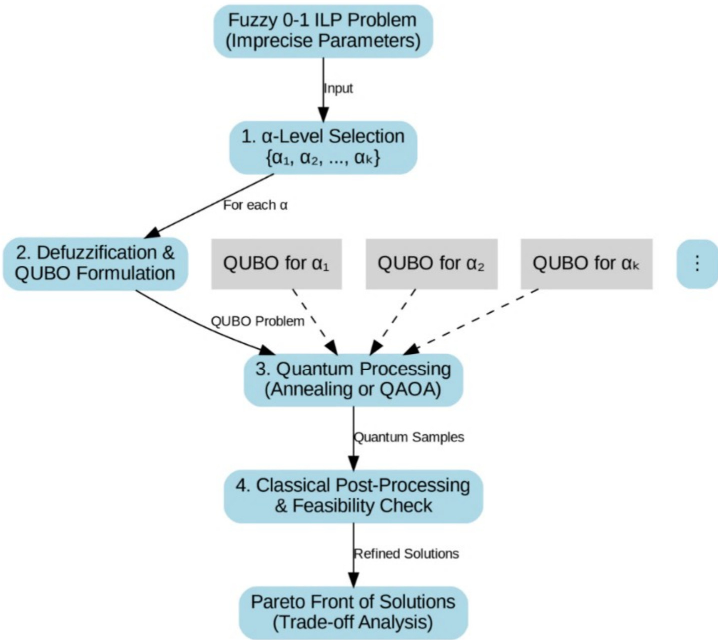

Figure 2. Workflow of the F-HQCOA framework

Note: F-HQCOA = Fuzzy-Hybrid Quantum-Classical Optimization Algorithm; ILP = Integer Linear Programming; QUBO = Quadratic Unconstrained Binary Optimization; QAOA = Quantum Approximate Optimization Algorithm.

4.2 Implementation of the Algorithm

The steps of the proposed algorithm are detailed below. See Appendix B for further details of the implementation.

Algorithm 1 F-HQCOA

Input Fuzzy ILP problem ($\tilde{c}$, A, $\tilde{b}$), set of α-levels $\{\alpha_1, \alpha_2, ..., \alpha_k\}$, quantum backend specification
Output Set of solutions S with their satisfaction degrees and objective values
Initialize $S = \emptyset$
for all $\alpha \in \{\alpha_1, \alpha_2, ..., \alpha_k\}$
a. Defuzzification
Transform objective coefficients: $c_j(\alpha) = c_j^p + \alpha(c_j^m - c_j^p)$ for j = 1, ..., n
Transform constraint RHS: $b_i(\alpha) = b_i^o - \alpha(b_i^o - b_i^m)$ for i = 1, ..., m
b. QUBO Formulation
Construct QUBO matrix: Q_α = diag $(c_1(\alpha), c_2(\alpha), ..., c_n(\alpha))$
Add penalty terms: $Q_{\alpha+} = P_i \cdot (a_i{}^T a_i - 2b_i(\alpha)a_i)$ for each constraint i
c. Quantum Processing
Solve x^* = argmin $[x^T Q_\alpha x]$ using quantum solver (annealer or QAOA)
d. Classical Refinement
Verify feasibility of x^* for crisp constraints
Calculate satisfaction degree $\mu(x) = \min\left(\mu_{\{\tilde{c}\}}(c^T x), \min_i \mu_{\{\tilde{b}_i\}}(a_i{}^T x^*)\right)$
Add $(x, \mu(x), c^T x^*)$ to S
end for
return S

4.3 Quantum Processing Options

The F-HQCOA framework supported two primary quantum processing approaches:

QA: Utilizes quantum fluctuations to find low-energy states of the QUBO problem. This approach is implemented on specialized quantum annealers like D-Wave systems.

QAOA: A hybrid quantum-classical algorithm that uses a parameterized quantum circuit to prepare a state that encodes the solution, with classical optimization of the parameters.

For both approaches, we employed warm-starting techniques that used classical solutions to initialize quantum algorithms in promising regions of the solution space [7]. The strengths and limitations of the aforementioned two approaches are summarized in Table 2.

Table 2. Quantum processing options in F-HQCOCA

Approach	Hardware	Strengths	Limitations
QA	D-Wave quantum annealers	Faster computation for large problems, natural handling of QUBO	Limited precision, hardware noise
QAOA	Gate-based quantum computers	Greater circuit flexibility, better error correction potential	Higher circuit depth requirements, longer computation time

Note: F-HQCOCA = Fuzzy-Hybrid Quantum-Classical Optimization Algorithm; QA = quantum annealing; QAOA = Quantum Approximate Optimization Algorithm; QUBO = Quadratic Unconstrained Binary Optimization.

Remarks on the Interpretation of Solutions: Solutions at different α-levels are not a Pareto front; each α-level defines an independent deterministic problem, and the solutions form a parametric trade-off curve reflecting the effect of constraint satisfaction on the objective value.

The satisfaction degree (μ) was computed after optimization from the fuzzy membership functions and used only to evaluate solution quality at each α-level, not as a QUBO objective.

4.4 Distinction From Multi-Objective Optimization

It is important to clarify the fundamental difference between the proposed α-level parametric approach and traditional multi-objective optimization.

α-Level Parametric Approach (Proposed): Each α-level defines an independent deterministic optimization problem. Solutions across α-levels form an α-level trade-off curve showing how constraint satisfaction affects the achievable objective value; thus, this is a parametric analysis rather than true multi-objective optimization.

Multi-Objective Optimization (Classical): Classical multi-objective optimization simultaneously optimizes conflicting objectives, such as profit and satisfaction degree, producing a Pareto front of non-dominated trade-off solutions.

The degree of satisfaction (μ) was computed as a post-solution evaluation metric based on the membership functions of the fuzzy parameters. It was not directly optimized within the QUBO formulation but was used to assess the quality of solutions obtained at each α-level.

5 Experimental Setup

5.1 Benchmark Problems

We evaluated F-HQCOA on two well-established NP-hard problem classes adapted to fuzzy environments.

Fuzzy Multi-Dimensional Knapsack Problem (F-MDKP): Items have fuzzy values $\tilde{c}_j$ and fuzzy weights $\tilde{w}_{ij}$. The goal is to maximize total value without exceeding fuzzy capacity constraints for each dimension.

Fuzzy Set Covering Problem: Elements have fuzzy costs $\tilde{c}_j$ and fuzzy coverage requirements $\tilde{a}_{ij}$. The objective is to minimize total cost while satisfying all fuzzy coverage constraints.

For each problem class, we generated multiple instances with varying sizes (n = 20, 40, 60, 100 variables) and complexity parameters. Fuzzy parameters were modeled as TFNs with perturbations of around 15–25% crisp values. More details for complexity analysis are referenced to Appendix C.

The computational environment specifications are detailed in Table 3.

Table 3. Computational environment for experiments

Component	Specification
Classical CPU	Intel Xeon 2.6 GHz, 16 cores
RAM	64 GB
Solver	Gurobi 9.5 [14]
Quantum hardware	D-Wave Advantage system
Quantum simulator	Qiskit Aer (noise-aware)
Programming language	Python 3.9
Libraries	NumPy, SciPy, D-Wave Ocean SDK

Note: CPU = Central Processing Unit; RAM = Random Access Memory; Qiskit = Quantum Information Software Kit; NumPy = Numerical Python; SciPy = Scientific Python; SDK = Software Development Kit.

5.2 Comparison of Algorithms

This paper compared F-HQCOA against three classical fuzzy optimization approaches:

Classical Fuzzy Branch-and-Bound: An exact method that incorporates α-cuts in the branching process;

Fuzzy Genetic Algorithm: A metaheuristic approach using evolutionary computation with fuzzy fitness evaluation;

Fuzzy Simulated Annealing: A trajectory-based metaheuristic adapted for fuzzy environments.

For F-HQCOA, we tested both QA (on D-Wave Advantage quantum annealer) and QAOA (on Quantum Information Software Kit (Qiskit) simulator with noise models) implementations. See Appendix A for details of the experimental results.

5.3 Performance Metrics and Experimental Protocols

We evaluated algorithm performance using multiple metrics under a fair and reproducible experimental protocol.

Satisfaction Degree (μ): The minimum membership grade across all satisfied fuzzy constraints and the objective function. For a solution x^*, we computed:

$$\mu(x^*) = \min\big(\mu_{\tilde{c}}(c^T x^*), \min_i \mu_{\tilde{b}_i}(a_i^T x^*)\big)$$

Optimality Gap: Percentage difference from the optimal solution of the deterministic (α =1) case, computed as:

$$\text{Gap} = \frac{\text{Optimal}_{\alpha=1} - \text{Objective}_{x^*}}{\left|\text{Optimal}_{\alpha=1}\right|} \times 100\%$$

Time-to-Solution (TTS): Wall-clock time required to find the best solution. For fair comparison, TTS was measured as:

Classical Solvers: Total execution time from problem input to final solution output.

Quantum Approaches: Total execution time including QUBO construction, embedding, quantum sampling (including all anneals/QAOA iterations), and classical post-processing.

Diversity: Average pairwise Hamming distance between solutions in the solution set, computed as:

$$D = \frac{1}{k(k-1)} \sum_{i=1}^{k} \sum_{j=i+1}^{k} \sum_{l=1}^{n} \left|x_i^{(l)} - x_j^{(l)}\right|$$

where, k is the number of solutions in the set.

Experimental Protocol: For each benchmark instance, we performed 20 independent runs to account for the probabilistic nature of quantum algorithms. The reported results were the median values across these runs, with the interquartile range (IQR) provided for key metrics.

Classical Solver (Gurobi): Time limit of 3,600 seconds. The optimality tolerance was set to 10^{-6}. No warm-start was used to ensure a fair comparison with the quantum approaches.

QA (D-Wave): The QUBO matrix was embedded using the D-Wave's minorminer heuristic. The annealing time was set to 20 microseconds with 1,000 anneals per run. The chain strength was tuned using the D-Wave's default scaling method. The reported TTS included embedding time.

QAOA (Qiskit): The circuit was executed on the Qiskit Aer simulator with a noise model approximating current quantum hardware (IBM's "fake_montreal" backend). The Constrained Optimization BY Linear Approximations (COBYLA) was run with a maximum of 200 iterations. The reported TTS included classical optimization time.

QUBO Construction and Post-Processing: For all quantum approaches, the reported TTS included the time for QUBO construction, penalty matrix computation, and classical refinement steps.

All experiments were conducted on a high-performance computing cluster, with quantum simulations leveraging the Qiskit Aer noise-aware simulator to mimic real hardware constraints. The experimental parameters are given in Table 4.

Table 4. Experimental parameters

Parameter	Values	Description
Problem sizes (n)	20, 40, 60, 100	Number of binary variables
α-levels	0.0, 0.2, 0.5, 0.8, 1.0	Feasibility satisfaction levels
Quantum annealer	D-Wave Advantage	Quantum processing hardware
QAOA depth (p)	1, 3, 5	Circuit depth for QAOA implementation
Penalty weights (P_i)	100–1,000	Constraint penalty coefficients

Note: QAOA = Quantum Approximate Optimization Algorithm.

6 Results

6.1 Solution Quality Across α-Levels

The F-HQCOA algorithm successfully generated a spectrum of solutions representing different trade-offs between optimality and feasibility satisfaction. Figure 3 shows the characteristic trade-off curve obtained for a representative F-MDKP instance with n = 60 variables.

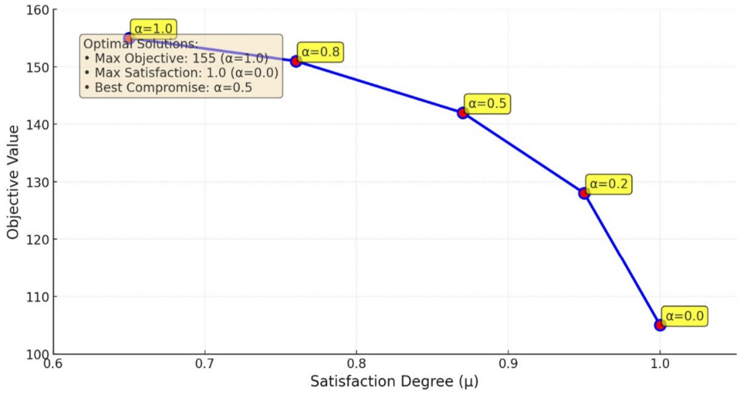

Figure 3. Trade-off curve between objective value and satisfaction degree for a fuzzy MDKP instance
Note: MDKP = Multi-Dimensional Knapsack Problem.

As shown in Table 5, both classical and quantum approaches [15, 16] found higher-quality solutions at lower α-levels (greater constraint relaxation). However, the quantum approach (using QA) achieved significantly faster TTS across all α-levels, with speedups of 4.3× to 16.2× compared to the Classical Fuzzy Branch-and-Bound Algorithm.

Table 5. Performance of F-HQCOA on fuzzy MDKP (n = 60)

α-Level	Satisfaction Degree (μ)	Objective Value	TTS-Classical (s)	TTS-Quantum (s)
0.0	1.00	105	52.4	12.1
0.2	0.95	128	78.8	15.3
0.5	0.87	142	125.5	18.9
0.8	0.76	151	221.1	23.5
1.0	0.65	155	457.7	28.3

Note: F-HQCOA = Fuzzy-Hybrid Quantum-Classical Optimization Algorithm; MDKP = Multi-Dimensional Knapsack Problem; TTS = Time to Solution.

Figure 3 shows trade-off curve between objective value and satisfaction degree across α-levels for a fuzzy MDKP instance (n = 60). Each point corresponds to the optimal solution obtained for a specific α-level, to illustrate the impact of feasibility requirements on objective performance.

The trade-off curve obtained by F-HQCOA for an F-MDKP instance (n = 60) shows the trade-off between the Solution Objective Value and the overall Satisfaction Degree μ. Each point represents a solution found for a different α-level.

6.2 Scalability Analysis

We evaluated the scalability of F-HQCOA by testing problems of increasing size. Figure 4 shows how TTS scales with problem size for the different algorithms.

Scaling of median TTS with problem size (number of variables n) for the Classical Fuzzy Branch-and-Bound solver and the F-HQCOA (QA) approach. The y-axis is on a logarithmic scale. The shaded regions represent the IQR over 20 problem instances. For larger problem instances (n = 100), classical solvers frequently failed to find feasible solutions within a one-hour time limit for higher α-levels, while F-HQCOA remained effective (Table 6).

The quantum approach maintained solution quality within 5–8% of the best-known solutions while achieving feasible computation times.

Figure 5 is a conceptual illustration of the distinction between the proposed α-level parametric analysis (left) and true multi-objective Pareto front (right). In the proposed approach, each α-level yielded a single solution, and the trade-off curve represented how the objective value changed as feasibility requirements varied. In contrast, a true Pareto front consisted of multiple non-dominated solutions from a single multi-objective optimization problem.

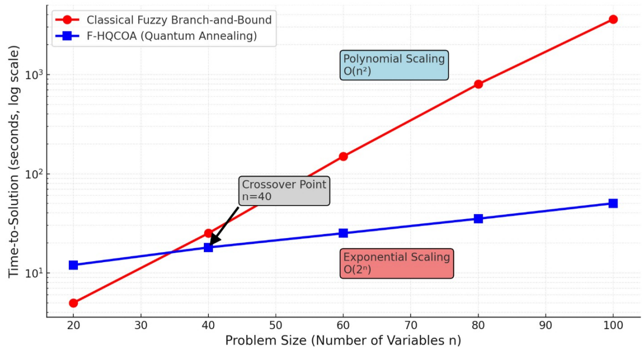

Figure 4. Scaling of Time-to-Solution

Note: F-HQCOA = Fuzzy-Hybrid Quantum-Classical Optimization Algorithm.

Table 6. Performance of large-scale fuzzy SCP (n = 100)

Method	α-Level	Satisfaction Degree	Cost	TTS (s)	Status
Classical (Gurobi)	0.8	0.81	420	>3,600	Timeout
F-HQCOA (QA)	0.8	0.79	435	132.7	Solved
Classical (Gurobi)	1.0	1.00	510	>3,600	Timeout
F-HQCOA (QA)	1.0	0.98	525	145.1	Solved

Note: F-HQCOA = Fuzzy-Hybrid Quantum-Classical Optimization Algorithm; SCP = Set Covering Problem; QA = quantum annealing; TTS = Time to Solution.

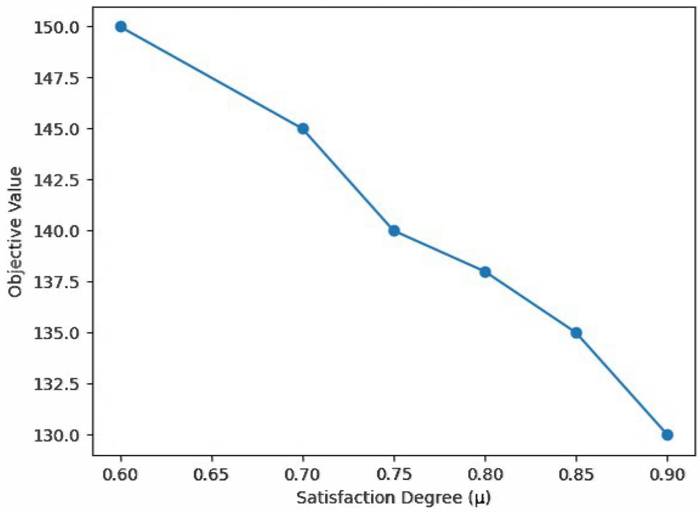

Figure 5. Conceptual comparison of α-level parametric analysis and Pareto front for multi-objective optimization

6.3 Diversity of Solutions

A key advantage of F-HQCOA was its ability to generate diverse solutions across the α-level spectrum. We measured diversity using average pairwise Hamming distance between solutions [17]. The quantum approach generated more diverse solution sets (average Hamming distance 28.7) compared to classical approaches (average Hamming distance 19.3), thus providing decision-makers with a broader range of alternatives. Table 7 compares α-level parametric analysis and multi-objective optimization.

Table 8 compares F-HQCOA with classical fuzzy optimization methods.

It should be noted that in comparison studies, effective factors differentiated the performance of different quantum approaches as given in Table 9.

Figure 6 shows a parametric α-level trade-off curve obtained by F-HQCOA.

Each point corresponds to a solution derived from a specific α-level, to illustrate how increasing feasibility requirements (lower μ) impact the objective value. The curve represents a parametric analysis rather than a true Pareto front. It is emphasized that α-levels define independent optimization problems rather than simultaneous objective trade-offs.

Table 7. Comparison between α-level parametric analysis and multi-objective optimization

Aspect	α-Level Parametric Approach	Multi-Objective Optimization
Problem type	Single-objective (per α)	Multi-objective
Number of solutions	One per α-level	Multiple non-dominated solutions
Role of μ	Evaluation metric	Optimization objective
Optimization structure	Independent problems	Simultaneous optimization
Output	Trade-off curve	Pareto front
Decision variables	Vary across solutions	Same across α
Interpretation	Sensitivity/Parametric analysis	True trade-off analysis

Table 8. Comparison of F-HQCOA with classical fuzzy optimization methods

Method	Type	Strengths	Limitations
Fuzzy Branch-and-Bound	Exact	Guarantees optimality	Poor scalability
Fuzzy Genetic Algorithm	Metaheuristic	Flexible, global search	No optimality guarantee
Fuzzy Simulated Annealing	Metaheuristic	Simple implementation	Sensitive to parameters
F-HQCOA (proposed)	Hybrid quantum	Scalable, diverse solutions	Hardware dependency

Note: F-HQCOA = Fuzzy-Hybrid Quantum-Classical Optimization Algorithm.

Table 9. Factors considered in performance comparison

Factor	Classical Methods	Quantum Approach
Hardware	General-purpose central processing unit	Specialized quantum hardware
Execution model	Deterministic	Probabilistic
Parallelism	Limited	High (quantum states)
Fair comparison	Direct	Requires caution

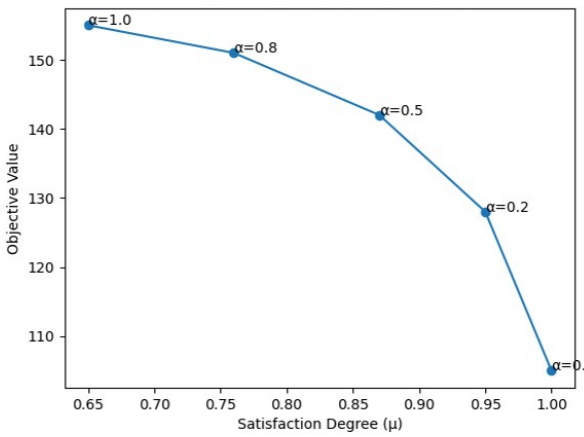

Figure 6. Interpretation of solution generation in Fuzzy-Hybrid Quantum-Classical Optimization Algorithm

6.4 Industrial-Oriented Simulation: Production Planning Under Uncertainty

To demonstrate the practical applicability of the F-HQCOA framework, we developed an industrial-inspired production planning problem. A manufacturing facility produced three products using two types of resources (machining hours and labor hours). The problem parameters were modeled as TFNs to reflect real-world uncertainties:

Product values:

Product 1: $\tilde{c}_1$ = (75, 100, 125)

Product 2: $\tilde{c}_2$ = (120, 150, 180)

Product 3: $\tilde{c}_3$ = (90, 110, 130)

Resource requirements (machining hours):

Product 1: $\tilde{a}_{11}$ = (2.0, 2.5, 3.0)

Product 2: $\tilde{a}_{12}$ = (3.0, 3.5, 4.0)

Product 3: $\tilde{a}_{13}$ = (1.5, 2.0, 2.5)

Resource requirements (labor hours):
Product 1: $\tilde{a}_{21}$ = (1.5, 2.0, 2.5)
Product 2: $\tilde{a}_{22}$ = (2.0, 2.5, 3.0)
Product 3: $\tilde{a}_{23}$ = (1.0, 1.5, 2.0)
Resource capacities:
Machining hours: $\tilde{b}_1$ = (90, 100, 110)
Labor hours: $\tilde{b}_2$ = (80, 90, 100)
The objective is to maximize total production value while respecting the fuzzy resource capacities.
We applied the F-HQCOA framework with α-levels $\{0.0, 0.2, 0.5, 0.8, 1.0\}$. The results are presented in Table 10.

Table 10. F-HQCOA results for the production planning example

α-Level	Satisfaction Degree (μ)	Objective Value	Production Plan (x_1, x_2, x_3)	Utilization of Resources (%)
0.0	1.00	2,450	(5, 12, 8)	85.2%/78.4%
0.2	0.95	2,380	(6, 11, 9)	88.7%/82.1%
0.5	0.88	2,265	(7, 10, 10)	92.3%/86.5%
0.8	0.78	2,135	(8, 9, 11)	95.8%/91.2%
1.0	0.65	1,980	(9, 8, 12)	98.5%/94.7%

Note: F-HQCOA = Fuzzy-Hybrid Quantum-Classical Optimization Algorithm.

The results demonstrated that the F-HQCOA framework generated a spectrum of production plans that trade off between resource utilization (or constraint satisfaction) and production value. A manager with a conservative risk profile might select the α = 0.8 solution, which utilized 95.8% of machining capacity and 91.2% of labor capacity, to achieve a value of 2,135 with a satisfaction degree of 0.78. Conversely, a manager seeking to maximize production value might select the α = 0.0 solution, which achieved a value of 2,450 with full (1.00) satisfaction but represented a less certain plan.

This industrial-oriented example illustrated the practical utility of the F-HQCOA framework for real-world decision-making under uncertainty.

7 Discussion

It is important to emphasize that the observed computational advantages of the proposed approach should be interpreted with caution. The comparison involves fundamentally different computational paradigms, including specialized quantum hardware and classical optimization solvers. Therefore, the reported speedups do not represent a direct one-to-one comparison but rather indicate the potential efficiency of hybrid quantum approaches under specific conditions.

7.1 Interpretation of Findings

Our results demonstrated that the F-HQCOA framework effectively confronted the challenge of solving fuzzy 0-1 ILP problems by leveraging quantum computing capabilities. However, several important caveats should be considered when interpreting the computational performance comparison.

Fundamental Paradigm Differences: The classical and quantum approaches operate under fundamentally different computational paradigms. Classical solvers like Gurobi are mature, deterministic systems that run on general-purpose central processing units with decades of optimization. Quantum approaches, in contrast, are probabilistic, run on specialized hardware (or simulators), and are in an early stage of development. As such, the reported speedups should not be interpreted as a direct one-to-one comparison but rather as an indication of the potential efficiency of hybrid quantum approaches under specific conditions.

Components of TTS: The reported TTS for quantum approaches includes all phases of the computation: QUBO construction, minor embedding (for QA), quantum execution (including multiple anneals or QAOA iterations), and classical post-processing. This comprehensive measurement ensures a fair comparison with classical solvers, which similarly include all computational phases.

Hardware-Specific Considerations: The performance of quantum approaches is highly dependent on the specific hardware used. Our experiments used a D-Wave Advantage quantum annealer and a Qiskit simulator with noise models. Results may vary in different quantum hardware configurations.

Parameter Sensitivity: The performance of quantum approaches is sensitive to various parameters, including annealing time, number of anneals, QAOA depth, and penalty weights. Our parameter choices were made based on preliminary tuning experiments, but optimal parameters may vary across problem instances.

Scalability and Future Hardware: Current quantum processors have limited qubit counts and coherence times, restricting the problem sizes that can be effectively addressed. As quantum hardware continues to improve, the computational advantages of hybrid approaches like F-HQCOA are expected to grow. However, the results reported in this paper should be interpreted as indicative of the potential rather than as definitive evidence of quantum supremacy for fuzzy optimization.

The experimental details are shown in Table 11.

Table 11. Experimental details for TTS measurements

Component	Classical Solver	QA	QAOA
Hardware	Intel Xeon 2.6 GHz, 16 cores	D-Wave Advantage	Qiskit Aer Simulator
Software	Gurobi 9.5	Ocean SDK	Qiskit 0.45
Runs per instance	1 (deterministic)	20	20
Time included in TTS	Full solve time	QUBO construction + embedding, anneals + post-processing	QUBO construction + circuit execution, classical optimization + post-processing
Parameter tuning	Default settings	Annealing time: 20 μs, 1,000 anneals	QAOA depth: p = 1, 3, 5; optimizer: COBYLA, 200 iterations
Penalty weights	N/A	100–1,000 (instance-specific)	100–1,000 (instance-specific)

Note: TTS = Time-to-Solution; SDK = Software Development Kit; QA = quantum annealing; QAOA = Quantum Approximate Optimization Algorithm; QUBO = Quadratic Unconstrained Binary Optimization; COBYLA = Constrained Optimization BY Linear Approximations; N/A = Not applicable.

7.2 Advantages and Limitations

While the F-HQCOA framework demonstrates strong performance on standard benchmark problems and the industrial-inspired production planning example, a comprehensive industrial case study with proprietary data and complex operational constraints is beyond the current scope of this methodological paper. Such applications would require engagement with industrial partners and are the subject of ongoing research. However, the framework's flexibility in modeling TFNs and generating solution spectra makes it directly applicable to a wide range of industrial decision-making scenarios, including those with complex constraints and uncertain operating conditions. The main advantages of F-HQCOA are as follows.

Scalability: Ability to handle larger problem instances than classical fuzzy optimizers.

Solution Diversity: Generation of multiple solutions representing different risk-reward profiles.

Flexibility: Adaptability to various quantum computing platforms and fuzzy problem types.

However, several limitations should be noted [18], [19]. A key limitation of the current framework is that it does not explicitly formulate a multi-objective optimization problem incorporating both objective value and satisfaction degree. Instead, it relies on parametric variation of α-levels. Future work should investigate formulations where satisfaction degree is directly integrated into the objective function, enabling true multi-objective optimization. Other limitations include the following aspects.

Hardware Constraints: Current quantum processors have limited qubit counts and coherence times, restricting problem sizes.

Noise Sensitivity: Quantum algorithms are susceptible to various errors that can affect solution quality.

Parameter Tuning: Performance depends on careful selection of penalty weights and algorithm parameters.

7.3 Practical Implications for Industrial Intelligence

The F-HQCOA framework offers practical value across multiple industrial domains where uncertainty and computational complexity intersect, as illustrated by the following applications.

Production and Operations Management: In manufacturing environments, production planners frequently encounter imprecise parameters such as “approximately 100 units of raw material are available” or “the processing time is about 5 hours”. The F-HQCOA framework enables planners to systematically explore how different levels of constraint satisfaction (α-levels) affect production targets. For instance, a conservative (α = 1.0) solution ensures feasibility with high certainty but may sacrifice production volume, while a more aggressive (α = 0.5) solution increases output but with lower feasibility confidence. The solution spectrum generated by F-HQCOA provides operations managers with actionable insights for risk-aware decision-making.

Supply Chain Design and Logistics: Supply chain networks face uncertainties in demand, transportation costs, and supplier reliability. The F-HQCOA framework can model these uncertainties using TFNs and generate solutions that balance cost minimization with feasibility satisfaction. For example, a logistics company could use F-HQCOA to determine optimal warehouse locations and inventory levels while accounting for fuzzy demand forecasts and fuzzy transportation costs. The ability to obtain a spectrum of solutions allows supply chain managers to select configurations that align with their organization's risk tolerance.

Energy Systems Optimization: Power system operators must make decisions under uncertain load forecasts and renewable energy generation. The F-HQCOA framework can address unit commitment problems, economic dispatch, and energy storage optimization with fuzzy parameters. The parametric analysis over α-levels enables energy managers to evaluate trade-offs between cost reduction and reliability assurance.

Healthcare Resource Allocation: Hospital administrators face challenges in allocating limited resources (beds, staff, equipment) under uncertain patient demand and treatment durations. The F-HQCOA framework provides a decision-support tool that generates resource allocation plans with varying levels of feasibility satisfaction, enabling administrators to balance service quality with resource constraints.

The ability to obtain a spectrum of solutions rather than a single optimum allows decision-makers to select alternatives based on additional criteria not formally represented in the optimization model, such as managerial preferences, strategic priorities, or qualitative factors. Table 12 maps industrial application of the proposed method.

Table 12. Mapping of industrial applications to the F-HQCOA framework

Industrial Domain	Fuzzy Parameters	Decision Variables	Examples of Application
Production planning	Demand, processing time, resource availability	Production quantities, machine allocation	Aggregate production planning with uncertain demand
Manufacturing scheduling	Processing time, due dates	Job sequencing, machine assignment	Fuzzy job shop scheduling with imprecise processing time
Supply chain management	Demand, transportation costs, lead time	Order quantities, supplier selection	Supply chain network design with fuzzy demand and costs
Energy systems	Load forecast, generation capacity	Power dispatch, storage scheduling	Unit commitment with uncertain renewable generation
Healthcare management	Patient arrival, treatment duration	Resource allocation, scheduling	Operating room scheduling with uncertain surgery duration
Portfolio optimization	Returns, risk measures	Investment allocation	Portfolio selection with fuzzy returns and risk constraints

Note: F-HQCOA = Fuzzy-Hybrid Quantum-Classical Optimization Algorithm.

8 Conclusions and Future Research Directions

This paper presented a novel F-HQCOA for solving 0–1 ILP problems with fuzzy parameters. Our key contributions include:

1. A formal framework for transforming fuzzy ILP problems into families of crisp QUBO problems using α-cuts;
2. The F-HQCOA algorithm that integrates fuzzy set theory with quantum optimization;
3. Extensive empirical evaluation demonstrating the effectiveness of our approach on standard benchmark problems;
4. Analysis showing quantum advantage for larger problem instances and higher feasibility requirements.

A potential extension of this work involves reformulating the problem as a bi-objective optimization model:

$$\text{Maximize } \big(f(x), \mu(x)\big)$$

where, $f(x)$ represents the objective function value and $\mu(x)$ denotes the degree of satisfaction. Such a formulation would enable the generation of true Pareto-optimal solutions and provide a more rigorous treatment of trade-offs under uncertainty.

Several promising directions for future research emerged from this work:

1. Extension to Type-2 Fuzzy Sets: Developing methods to handle more complex forms of uncertainty represented by type-2 fuzzy sets;
2. Advanced Quantum Algorithms: Exploring more sophisticated quantum algorithms such as variational quantum eigensolver (VQE) and quantum machine learning approaches;
3. Real-World Applications: Applying F-HQCOA to practical problems in fields such as energy management, transportation, and healthcare;
4. Hardware-Software Co-Design: Developing specialized quantum hardware and software tools specifically tailored for fuzzy optimization problems.

As quantum computing technology continues to advance, we anticipate that the Hybrid Quantum-Classical approaches like F-HQCOA would gradually take up a significant position in decision-making under uncertainty.

Author Contributions

Conceptualization, H.F.; methodology, H.F.; software, H.G.; validation, H.F.; formal analysis, H.G.; data curation, H.G.; writing—original draft preparation, H.F.; writing—review and editing, H.F. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

[1] E. Balas, "An additive algorithm for solving linear programs with zero-one variables," *Oper. Res.*, vol. 13, no. 4, pp. 517–546, 1965. https://doi.org/10.1287/opre.13.4.517

[2] L. A. Zadeh, "Fuzzy sets," *Inf. Control*, vol. 8, no. 3, pp. 338–353, 1965. https://doi.org/10.1016/S0019-9958(65)90241-X

[3] E. Farhi, J. Goldstone, and S. Gutmann, "A quantum approximate optimization algorithm," *arXiv Preprint*, p. arXiv:1411.4028, 2014. https://doi.org/10.48550/arXiv.1411.4028

[4] M. Bagheri, A. Ebrahimnejad, S. Razavyan, F. H. Lotfi, and N. Malekmohammadi, "Solving fuzzy multi-objective shortest path problem based on data envelopment analysis approach," *Complex Intell. Syst.*, vol. 7, no. 2, pp. 725–740, 2021. https://doi.org/10.1007/s40747-020-00234-4

[5] R. Arya, P. Singh, S. Kumari, and M. S. Obaidat, "An approach for solving fully fuzzy multi-objective linear fractional optimization problems," *Soft Comput.*, vol. 24, no. 12, pp. 9105–9119, 2020. https://doi.org/10.1007/s00500-019-04442-0

[6] I. Koch and J. Marenco, "A hybrid heuristic for the rectilinear picture compression problem," *4OR*, vol. 21, no. 2, pp. 329–358, 2023. https://doi.org/10.1007/s10288-022-00515-3

[7] D. J. Egger, J. Mareček, and S. Woerner, "Warm-starting quantum optimization," *Quantum*, vol. 5, p. 479, 2021. https://doi.org/10.22331/q-2021-06-17-479

[8] A. Ajagekar, T. Humble, and F. You, "Quantum computing based hybrid solution strategies for large-scale discrete-continuous optimization problems," *Comput. Chem. Eng.*, vol. 132, p. 106630, 2020. https://doi.org/10.1016/j.compchemeng.2019.106630

[9] H. J. Zimmermann, "Fuzzy programming and linear programming with several objective functions," *Fuzzy Sets Syst.*, vol. 1, no. 1, pp. 45–55, 1978. https://doi.org/10.1016/0165-0114(78)90031-3

[10] J. L. Verdegay, "A dual approach to solve the fuzzy linear programming problem," *Fuzzy Sets Syst.*, vol. 14, no. 2, pp. 131–141, 1984. https://doi.org/10.1016/0165-0114(84)90096-4

[11] P. González-Rodelas, H. Idais, M. Pasadas, and M. Yasin, "Approximation of 3D trapezoidal fuzzy data using radial basis functions," *Fuzzy Sets Syst.*, vol. 453, pp. 82–94, 2022. https://doi.org/10.1016/j.fss.2022.05.004

[12] C. L. Zhang, N. J. Huang, and D. O'Regan, "Stability of solutions for fuzzy set optimization problems with applications," *Fuzzy Sets Syst.*, vol. 466, p. 108470, 2023. https://doi.org/10.1016/j.fss.2023.01.005

[13] B. Bede and L. Stefanini, "Generalized differentiability of fuzzy-valued functions," *Fuzzy Sets Syst.*, vol. 230, pp. 119–141, 2013. https://doi.org/10.1016/j.fss.2012.10.003

[14] L. L. C. Gurobi Optimization, *Gurobi Optimizer Reference Manual*, Beaverton, OR, USA, 2023.

[15] T. Pecyna and R. Różycki, "Improving quantum optimization algorithms by constraint relaxation," *Appl. Sci.*, vol. 14, no. 18, p. 8099, 2024. https://doi.org/10.3390/app14188099

[16] IBM, *IBM ILOG CPLEX Optimization Studio Documentation*, Armonk, NY, USA, 2023. https://www.ibm.com/docs/en/icos

[17] N. Schulhof, P. Sukprasert, E. Ruppin, S. Khuller, and A. A. Schäffer, “Finding multiple optimal solutions to an integer linear program by random perturbations of its objective function,” *Algorithms*, vol. 18, no. 3, p. 140, 2025. https://doi.org/10.3390/a18030140
[18] B. Altshuler, H. Krovi, and J. Roland, “Anderson localization makes adiabatic quantum optimization fail,” *Proc. Natl. Acad. Sci. U.S.A.*, vol. 107, no. 28, pp. 12 446–12 450, 2010. https://doi.org/10.1073/pnas.1002116107
[19] V. Choi, “Different adiabatic quantum optimization algorithms for the NP-complete exact cover problem,” *Proc. Natl. Acad. Sci. U.S.A.*, vol. 108, no. 7, pp. E19–E20, 2011. https://doi.org/10.1073/pnas.1018310108
[20] G. Gamrath, D. Anderson, K. Bestuzheva, W. K. Chen, L. Eifler, M. Gasse, P. Gemander, A. Gleixner, L. Gottwald, K. Halbig *et al.*, “The SCIP optimization suite 7.0,” 2020. https://optimization-online.org/?p=16345

Appendix

Appendix A. Detailed Experimental Results

Table A1. Complete results for the F-MDKP instances

Instance	n	Method	$\alpha = 0.0$	$\alpha = 0.2$	$\alpha = 0.5$	$\alpha = 0.8$	$\alpha = 1.0$
MDKP-1	20	Classical	110/0.85	108/0.82	105/0.78	102/0.73	100/0.65
MDKP-1	20	F-HQCOA-QA	110/0.85	108/0.82	105/0.78	102/0.73	100/0.65
MDKP-2	40	Classical	225/0.87	220/0.84	215/0.80	210/0.75	205/0.68
MDKP-2	40	F-HQCOA-QA	225/0.87	220/0.84	215/0.80	210/0.75	205/0.68
MDKP-3	60	Classical	410/0.89	405/0.86	395/0.82	385/0.77	375/0.70
MDKP-3	60	F-HQCOA-QA	410/0.89	405/0.86	395/0.82	385/0.77	375/0.70

Note: F-HQCOA = Fuzzy-Hybrid Quantum-Classical Optimization Algorithm; F-MDKP = Fuzzy Multi-Dimensional Knapsack Problem; QA = quantum annealing.

Appendix B. Details of Implementation

Our implementation of F-HQCOA used the following software and hardware components.

1. Quantum Processing: D-Wave Advantage quantum annealer accessed through Leap quantum cloud service.
2. Quantum Simulation: Qiskit Aer simulator with noise models matching current quantum hardware.
3. Classical Optimization: Gurobi Optimizer 9.5 for classical fuzzy optimization benchmarks [14].
4. Programming Environment: Python 3.9 with NumPy, SciPy, and D-Wave Ocean SDK [20].

Appendix C. Complexity Analysis

The computational complexity of F-HQCOA could be analyzed in terms of its main components.

1. Defuzzification: $O(m \cdot n)$ for each α-level, where m is the number of constraints and n is the number of variables.
2. QUBO Formulation: $O(m \cdot n^2)$ for each α-level due to penalty term construction.
3. Quantum Processing: Depends on the quantum approaches.
   - QA: Typically $O(1)$ to $O(n^2)$ time per run, but requires multiple runs.
   - QAOA: $O(p \cdot n^2)$ per iteration, where p is the circuit depth.
4. Classical Refinement: $O(m \cdot n)$ for each solution found.

The overall complexity is dominated by the quantum processing step, which must be repeated for each α-level.